INCH-POUND

MS90537B
7 September 2007
SUPERSEDING
MS90537A
7 March 1973

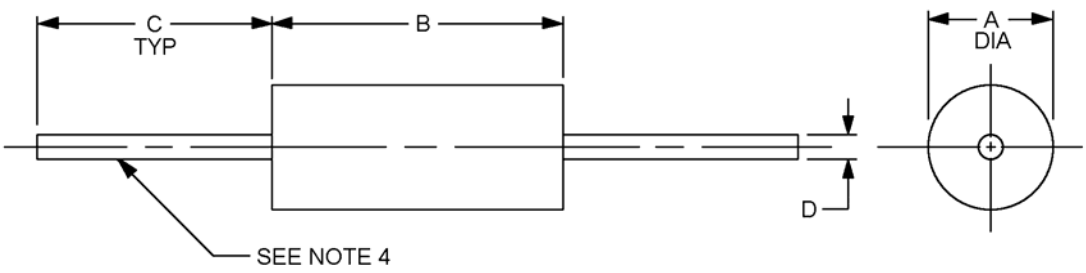
MILITARY SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, MOLDED, SUBMINIATURE, SHIELDED, MAGNETIC, TYPES LT4K242 TO LT4K314

Inactive for new design, after 7 March 1973.
For new design, use MS75087, MS75088 and MS75089.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.




| LTR | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Dimensions | Tolerance |
| A | .157 (3.99) | ±.010 (±0.25) |
| B | .395 (10.03) | ±.020 (±0.51) |
| C | 1.438 (36.53) | ±.188 (±4.78) |
| D | .025 (0.64) | ±.002 (±0.05) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their bodies.
4. Solderable/weldable lead wire, tinned copper wire, number 22 AWG.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT4

Grade: 1
Class: B

Weight: .750 grams or less.

Maximum operating temperature: +125°C.

Operating temperature range: -55°C to +125°C.

Ambient temperature: +90°C ±5°C.

Temperature rise: 35°C, maximum.

Terminal pull: 5 pounds, minimum.

Altitude: 70,000 feet.

Shock, specified pulse: Method 213 of MIL-STD-202, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: Method 301 of MIL-STD-202, test voltage 700 V rms.

At reduced barometric pressure: Method 105 of MIL-STD-202, test condition C, test voltage 180 V rms.

Percent coupling: 3 percent, maximum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS90537 - (dash number from table I).



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TABLE I. Electrical characteristics (initial).

| Dash Number MS90537 | Inductance (μH) ±10% | Q Min. | Test frequency (MHz) | SRF Minimum (MHz) | DC resistance max (Ohms) | 1/ Rated DC current, (mA) | Incremental current (mA) |
|---|---|---|---|---|---|---|---|
| -1 | 0.10 | 50 | 25 | 250 | .025 | 2900 | 2900 |
| -2 | 0.12 | 51 | 25 | 250 | .034 | 2800 | 2800 |
| -3 | 0.15 | 51 | 25 | 250 | .037 | 2750 | 2750 |
| -4 | 0.18 | 50 | 25 | 250 | .047 | 2200 | 2200 |
| -5 | 0.22 | 49 | 25 | 250 | .067 | 1700 | 1700 |
| -6 | 0.27 | 47 | 25 | 250 | 0.11 | 1500 | 1500 |
| -7 | 0.33 | 46 | 25 | 250 | 0.13 | 1300 | 1300 |
| -8 | 0.39 | 44 | 25 | 250 | 0.18 | 1100 | 1100 |
| -9 | 0.47 | 44 | 25 | 235 | 0.25 | 1000 | 1000 |
| -10 | 0.56 | 43 | 25 | 210 | 0.33 | 900 | 900 |
| -11 | 0.68 | 42 | 25 | 190 | 0.45 | 750 | 750 |
| -12 | 0.82 | 40 | 25 | 180 | 0.59 | 600 | 600 |
| -13 | 1.00 | 47 | 25 | 140 | 0.07 | 1900 | 1900 |
| -14 | 1.20 | 46 | 7.9 | 130 | .093 | 1600 | 1600 |
| -15 | 1.50 | 45 | 7.9 | 115 | 0.12 | 1300 | 1300 |
| -16 | 1.80 | 43 | 7.9 | 105 | 0.14 | 1200 | 1200 |
| -17 | 2.20 | 45 | 7.9 | 100 | 0.19 | 1100 | 1100 |
| -18 | 2.70 | 46 | 7.9 | 92 | 0.28 | 950 | 950 |
| -19 | 3.30 | 44 | 7.9 | 85 | 0.35 | 800 | 800 |
| -20 | 3.90 | 44 | 7.9 | 75 | 0.40 | 750 | 750 |
| -21 | 4.70 | 44 | 7.9 | 70 | 0.55 | 650 | 650 |
| -22 | 5.60 | 47 | 7.9 | 65 | 0.72 | 550 | 550 |
| -23 | 6.80 | 50 | 7.9 | 55 | 1.02 | 500 | 500 |
| -24 | 8.20 | 50 | 7.9 | 50 | 1.32 | 475 | 475 |
| -25 | 10.0 | 49 | 7.9 | 46 | 1.62 | 450 | 450 |
| -26 | 12.0 | 55 | 2.5 | 44 | 2.00 | 400 | 400 |
| -27 | 15.0 | 44 | 2.5 | 49 | 0.80 | 620 | 250 |
| -28 | 18.0 | 45 | 2.5 | 45 | 0.89 | 610 | 235 |
| -29 | 22.0 | 46 | 2.5 | 41 | 0.96 | 600 | 220 |
| -30 | 27.0 | 49 | 2.5 | 38 | 1.19 | 500 | 200 |
| -31 | 33.0 | 45 | 2.5 | 34 | 1.37 | 490 | 190 |
| -32 | 39.0 | 53 | 2.5 | 29 | 1.93 | 410 | 180 |
| -33 | 47.0 | 52 | 2.5 | 27 | 2.11 | 400 | 175 |
| -34 | 56.0 | 49 | 2.5 | 25 | 2.23 | 380 | 160 |
| -35 | 68.0 | 51 | 2.5 | 21 | 2.70 | 370 | 150 |
| -36 | 82.0 | 45 | 2.5 | 10.5 | 2.44 | 360 | 140 |
| -37 | 100.0 | 52 | 2.5 | 10.0 | 3.12 | 325 | 120 |
| -38 | 120.0 | 57 | .79 | 9.7 | 3.60 | 290 | 95 |
| -39 | 150.0 | 56 | .79 | 8.5 | 4.10 | 275 | 90 |
| -40 | 180.0 | 60 | .79 | 8.0 | 4.40 | 260 | 85 |
| -41 | 220.0 | 58 | .79 | 7.5 | 5.00 | 250 | 80 |
| -42 | 270.0 | 60 | .79 | 7.0 | 5.80 | 240 | 70 |
| -43 | 330.0 | 54 | .79 | 6.5 | 6.40 | 225 | 65 |
| -44 | 390.0 | 67 | .79 | 6.2 | 7.40 | 200 | 60 |
| -45 | 470.0 | 60 | .79 | 5.7 | 9.50 | 180 | 58 |
| -46 | 560.0 | 60 | .79 | 4.7 | 10.5 | 174 | 55 |
| -47 | 680.0 | 60 | .79 | 4.5 | 11.8 | 168 | 50 |
| -48 | 820.0 | 57 | .79 | 4.2 | 13.0 | 152 | 45 |
| -49 | 1,000.0 | 65 | .79 | 3.8 | 17.5 | 135 | 40 |
| -50 | 1,200.0 | 45 | .25 | 1.5 | 22.1 | 115 | 35 |

See footnote at end of table.



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TABLE I. Electrical characteristics (initial) – Continued.

| Dash Number MS90537 | Inductance (μH) ±10% | Q Min. | Test frequency (MHz) | SRF Minimum (MHz) | DC resistance max (Ohms) | 1/ Rated DC current, (mA) | Incremental current (mA) |
|---|---|---|---|---|---|---|---|
| -51 | 1,500.0 | 49 | .25 | 1.2 | 26.5 | 110 | 33 |
| -52 | 1,800.0 | 47 | .25 | 1.0 | 29.9 | 105 | 30 |
| -53 | 2,200.0 | 50 | .25 | 0.97 | 33.8 | 99 | 27 |
| -54 | 2,700.0 | 47 | .25 | 0.92 | 47.3 | 83 | 25 |
| -55 | 3,300.0 | 43 | .25 | 0.84 | 53.0 | 80 | 22 |
| -56 | 3,900.0 | 43 | .25 | 0.80 | 73.8 | 67 | 20 |
| -57 | 4,700.0 | 44 | .25 | 0.74 | 81.6 | 63 | 19 |
| -58 | 5,600.0 | 45 | .25 | 0.73 | 98.9 | 56 | 17 |
| -59 | 6,800.0 | 43 | .25 | 0.66 | 111.0 | 54 | 16 |
| -60 | 8,200.0 | 42 | .25 | 0.54 | 119.0 | 52 | 15 |
| -61 | 10,000.0 | 39 | .25 | 0.47 | 137.0 | 49 | 14 |
| -62 | 12,000.0 | 31 | .079 | 0.33 | 143.0 | 46 | 13 |
| -63 | 15,000.0 | 31 | .079 | 0.29 | 157.0 | 45 | 12 |
| -64 | 18,000.0 | 31 | .079 | 0.28 | 225.0 | 41 | 10 |
| -65 | 22,000.0 | 27 | .079 | 0.25 | 274.0 | 33 | 9 |
| -66 | 27,000.0 | 27 | .079 | 0.21 | 308.0 | 31 | 8 |
| -67 | 33,000.0 | 27 | .079 | 0.19 | 343.0 | 30 | 7.5 |
| -68 | 39,000.0 | 27 | .079 | 0.17 | 376.0 | 27 | 6.0 |
| -69 | 47,000.0 | 23 | .079 | 0.16 | 473.0 | 26 | 5.5 |
| -70 | 56,000.0 | 23 | .079 | 0.14 | 512.0 | 25 | 5.0 |
| -71 | 68,000.0 | 23 | .079 | 0.13 | 580.0 | 24 | 4.0 |
| -72 | 82,000.0 | 21 | .079 | 0.12 | 618.0 | 23 | 3.5 |
| -73 | 100,000.0 | 18 | .079 | 0.11 | 678.0 | 22 | 3.0 |

1/ For the overload test, direct current shall be used. The value of this current shall be 1.5 times the rated current.

TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from Initial measurement | | Allowable percent from specified minimum value in electrical characteristics (initial) table | |
|---|---|---|---|---|
| | Inductance (percent) | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | | | | |
| Group II | ±5 | --- | --- | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -5 | -10 |
| Conformance inspection group C | | | | |
| Subgroup I | ±5 | --- | --- | -10 |
| Subgroup II | ±5 | ±(2% +.001 ohm) | -5 | -10 |



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Application Notes:

1. Life test is not applicable.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-15305
MIL-STD-202

Custodians:
Army – CR
Navy - EC
Air Force - 11
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2007-044

Review activities:
Army – MI
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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